Bifurcation-structured thermo–swelling oscillations in a thermo-responsive gel: a minimal model and dynamical phase diagram

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Autonomous thermo–swelling oscillations in internally heated LCST gels arise from feedback between temperature-dependent collapse and collapse-suppressed heat input, but the bifurcation structures that delimit stable large-amplitude oscillations remain unclear. We study a minimal two-variable model coupling overdamped swelling kinetics to a lumped heat balance. We find that, in the heating–cooling parameter plane, the stable oscillation region is bounded by global bifurcations of periodic orbits rather than by the Hopf curve: the Hopf bifurcation is subcritical in the explored regime and marks only the loss of fixed-point stability. This global organization produces a broad region of fixed-point/limit-cycle coexistence. We further interpret the dynamics in a fast–slow framework and quantify how the oscillation period and waveform vary with heating, cooling, and feedback sharpness.

I. INTRODUCTION

Thermo-responsive gels provide a simple setting in which nonlinear mechanics, phase-transition-like volume changes, and energy exchange can combine to produce autonomous motion [1–5]. In LCST-type networks, increasing temperature reduces solvent quality and drives collapse, while cooling restores swelling [1, 2, 6]. When a gel is subject to sustained internal heating and simultaneous environmental cooling, and when the collapsed state suppresses the heat input, these ingredients close a negative thermo-swelling feedback loop capable of generating self-excited dynamics [7–11]. In this class of systems, the oscillatory mechanism does not require an intrinsically oscillatory chemical reaction; rather, it arises from the coupling between temperature-dependent swelling, hysteretic volume change, and feedback-controlled heating [7–10, 12].

Autonomous oscillations in gels have been demonstrated and modeled in several contexts, including chemically driven oscillatory materials and stimulus-coupled responsive polymers [7–11, 13–15]. For thermo-swelling oscillators, however, the theoretical picture remains incomplete. High-dimensional descriptions often retain spatial transport, poroelasticity, or large-deformation diffusion mechanics [16–18], whereas reduced models frequently focus on time-domain simulations, mechanism illustration, or local onset conditions rather than on a complete bifurcation organization of stable oscillations [8, 15, 19]. What remains unclear is the experimentally relevant question: in a thermo-swelling oscillator governed by LCST collapse and collapse-suppressed heating, which bifurcation structures actually determine the existence domain of stable large-amplitude oscillations?

This question is nontrivial because the swelling subsystem already possesses a nonlinear equilibrium manifold with fold-induced hysteresis, while the thermal balance introduces a second variable and a feedback-controlled energy input. In such a setting, the onset of oscillatory instability at a fixed point need not coincide with the creation or destruction of the attracting relaxation cycle observed in practice. A local Hopf calculation can therefore be insufficient to delimit the stable oscillatory regime. What is needed is a systematic bifurcation analysis that separates local fixed-point instability from the global periodic-orbit geometry that organizes stable self-sustained motion.

In this paper, we address that issue using a minimal spatially lumped two-variable model for an internally heated thermo-responsive gel. The model couples overdamped swelling kinetics to a lumped heat balance, with LCST solvent-quality changes providing the swelling nonlinearity [2, 6, 20] and a collapse-suppressed heating gate closing the feedback loop [7–10, 12]. Within this framework, we determine how the heating and cooling parameters organize fixed points, relaxation oscillations, and coexistence regimes. Our central result is that, in the explored parameter window, the Hopf bifurcation is subcritical and marks only the loss of fixed-point stability, whereas the stable oscillation domain is delimited by global bifurcations of periodic orbits. This global organization produces a broad region of fixed-point/limit-cycle coexistence and provides a natural explanation for the large-amplitude relaxation cycles seen in the model.

II. MODEL

We seek a minimal, spatially lumped "reaction–heat–swelling" model for a thermo-responsive hydrogel. The gel is assumed to remain spatially uniform, so its state is fully characterized by the polymer volume fraction $\phi(t) \in (0, 1)$ and the (spatially averaged) gel temperature $T(t)$.

We follow the gel-lattice-spring/Flory–Huggins formulation [20, 21] in which the free-energy density (per reference volume) is written as $U(\phi, T) = U_{\mathrm{el}} + U_{\mathrm{FH}}$. The elastic part is taken as neo-Hookean,

$$U_{\mathrm{el}} = \frac{c_0 v_0}{2} \left(I_1 - 3 - \ln I_3 \right), \tag{1}$$

where c_0 is the crosslink density, v_0 is the monomeric volume, and $I_1 = \mathrm{tr}\,\hat{\mathbf{B}}$ and $I_3 = \det \hat{\mathbf{B}}$ are invariants of

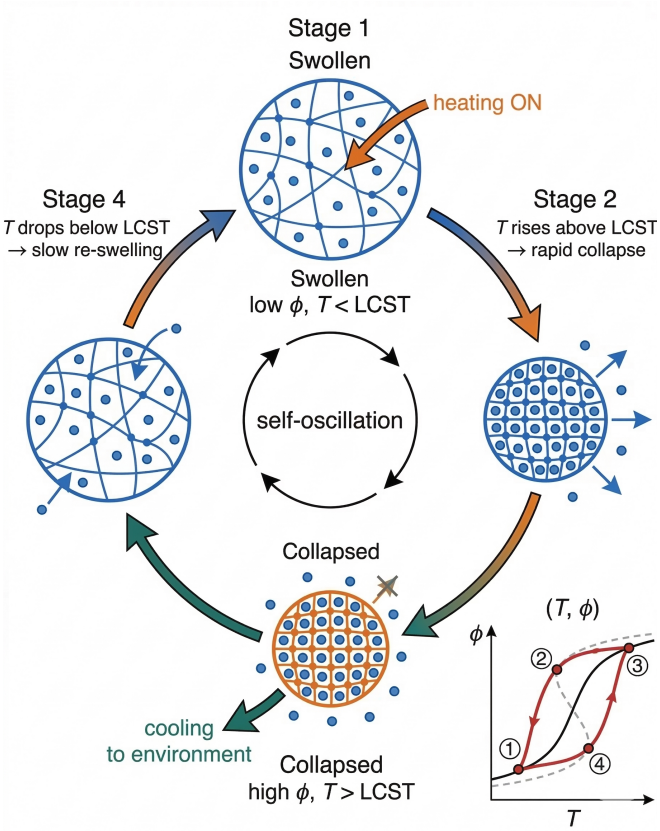

FIG. 1. Schematic of the thermo-swelling self-oscillation mechanism. Internal heating drives gel collapse; collapse suppresses the heat input; environmental cooling restores swelling, closing the negative feedback loop that sustains autonomous oscillations.

the left Cauchy–Green tensor $\hat{\mathbf{B}} = \hat{\mathbf{F}}\hat{\mathbf{F}}^{\mathsf{T}}$. Unless otherwise noted, stresses and pressures are nondimensionalized by $k_B T_0/v_0$, with T_0 the ambient temperature; the combination $c_0 v_0$ in Eq. (1) acts as a dimensionless elastic modulus calibrated at T_0, and temperature-dependence of rubber elasticity is neglected at this minimal level. The Flory–Huggins mixing contribution is [20, 21]

$$U_{\mathrm{FH}} = I_3^{1/2}\Big[(1-\phi)\ln(1-\phi) + \chi(\phi,T)\,\phi(1-\phi)\Big], \quad (2)$$

where $\chi(\phi,T)$ is the polymer–solvent interaction parameter and the prefactor $I_3^{1/2}$ converts from deformed to reference volume.

In the lumped approximation the deformation is isotropic. Polymer incompressibility implies $\phi = \phi_0/J$ with $J = \det\hat{\mathbf{F}}$, and for isotropic swelling one has $I_3^{1/2} = J = \phi_0/\phi$. The elastic contribution then reduces to an effective (dimensionless) elastic pressure

$$\Pi_{\mathrm{el}}(\phi) = c_0 v_0 \left[\left(\frac{\phi}{\phi_0}\right)^{1/3} - \frac{\phi}{2\phi_0}\right], \quad (3)$$

and we adopt the following reduced form for the mixing (osmotic) pressure,

$$\pi_{\mathrm{osm}}(\phi,T) = -\Big[\phi + \ln(1-\phi) + \chi(\phi,T)\,\phi^2\Big]. \quad (4)$$

When χ depends explicitly on ϕ, a fully free-energy-consistent derivation of the osmotic pressure generally generates additional contributions proportional to $\partial\chi/\partial\phi$. Here we follow the standard constitutive closure used in Flory-type gel modeling and treat Eq. (4) as an effective osmotic pressure within this minimal two-variable description [20, 22]. Swelling/deswelling is overdamped: in the two-fluid picture, polymer–solvent friction balances thermodynamic forces. For a spatially uniform gel of characteristic size L, this leads to a single relaxational timescale τ_ϕ (collective diffusion time) and the lumped kinetic law [18, 23]

$$\frac{d\phi}{dt} = \frac{1}{\tau_\phi}\Big[\Pi_{\mathrm{el}}(\phi) - \pi_{\mathrm{osm}}(\phi,T)\Big], \quad (5)$$

expressing the competition between elastic retraction and osmotic swelling.

For LCST-type gels such as pNIPAAm, solvent quality decreases with increasing temperature. We adopt the standard affine form [2, 6, 20]

$$\chi(\phi,T) = \chi_0(T) + \chi_1\phi, \qquad \chi_0(T) = \frac{\Delta h - T\Delta s}{k_B T}, \quad (6)$$

where T is the absolute temperature (K), $\Delta h < 0$ and $\Delta s < 0$ are the mixing enthalpy and entropy per monomer, k_B is Boltzmann's constant, and χ_1 captures concentration dependence. This form produces LCST behavior because χ increases with T [2, 6].

The gel is internally heated—by photothermal absorption or an exothermic reaction—and exchanges heat with the environment [7–10, 12, 24]. Assuming spatial uniformity (lumped capacitance), the energy balance gives

$$\rho_m c_p V\,\frac{dT}{dt} = (-\Delta H)\,V\,r - hA\,(T - T_0), \quad (7)$$

where $-\Delta H > 0$ is the exothermicity per reacted amount, r is the volumetric reaction rate, h is the heat-transfer coefficient, A is the exchange area, and T_0 is the ambient temperature. Defining the effective cooling rate $k \equiv hA/(\rho_m c_p V)$, we note that in principle V and A vary with swelling; to keep the model minimal we absorb such geometric dependence into the effective constant k. We do not model explicit reaction dynamics or reactant depletion; instead, the term $\gamma f(\phi)$ is a phenomenological power input that represents collapse-induced suppression of heat generation (e.g., reduced reactant transport, decreased optical-absorption cross section upon deswelling), with $\gamma \equiv (-\Delta H)r_0/(\rho_m c_p)$ a heating rate and $f(\phi) \in (0,1]$ a dimensionless gate [8–10, 12]. This closure yields

$$\frac{dT}{dt} = \gamma f(\phi) - k(T - T_0), \qquad \gamma \equiv \frac{(-\Delta H)r_0}{\rho_m c_p}. \quad (8)$$

TABLE I. Fixed material parameters used throughout. τ_ϕ sets the swelling timescale; γ and k are the primary control parameters explored in Sec. III.

Parameter	Value	Meaning
ϕ_0	0.139	reference volume fraction
$c_0 v_0$	3.81×10^{-3}	dimensionless elastic modulus
χ_1	0.908	concentration-coupling coefficient
Δh	-4.59×10^{-21} J	mixing enthalpy per monomer
Δs	-1.98×10^{-23} J K^{-1}	mixing entropy per monomer
τ_ϕ	0.1 s	swelling relaxation time
T_0	298.15 K	ambient temperature (25 °C)
ϕ_g	0.139	gate threshold
κ	30	gate steepness coefficient
n	3	Hill exponent (baseline)

To represent the sharp suppression of heat generation upon collapse, we choose a Hill-type gate [8–10, 15]

$$f(\phi) = \frac{1}{1 + \Big[\kappa \max(\phi - \phi_g, 0)\Big]^n}, \tag{9}$$

where ϕ_g is the collapse threshold, $\kappa > 0$ controls steepness, and $n \geq 1$ controls sharpness. The $\max(\cdot, 0)$ ensures $f \equiv 1$ for $\phi \leq \phi_g$ (swollen, maximal heating) and $f \to 0$ as $\phi \to 1$ (collapsed, heating suppressed), keeping $f \in (0, 1]$ for all ϕ. For $n \geq 2$ one has $f'(\phi_g^+) = 0$, so $f(\phi)$ is C^1 at the threshold and the Jacobian-based stability analysis is well-defined. The baseline value $n = 3$ gives a sigmoidal switch consistent with the nonlinear network response [8, 15]; the dependence on n is explored below.

Equations (5) and (8), together with (3)–(4), (6), and (9), define the closed two-variable system for $(\phi(t), T(t))$ studied in this paper. The feedback loop is: swollen gel (ϕ small) $\Rightarrow f \approx 1$, strong heating $\Rightarrow$ elevated T increases χ and drives osmotic collapse $\Rightarrow f \approx 0$, cooling dominates $\Rightarrow$ reduced T restores solvent affinity and gel re-swells. This loop underlies the relaxation oscillations analyzed in Sec. III.

The baseline parameters (Table I) are taken from experimental data for NIPAAm gels [2, 6]: $\phi_0 = 0.139$ and $c_0 v_0 = 3.81 \times 10^{-3}$ follow Ref. [6]; Δh, Δs, and χ_1 reproduce the LCST transition near 32 °C.

III. STABILITY AND BIFURCATION ANALYSIS

A. Equilibrium manifold and fixed points

In the absence of thermal feedback ($\dot{T} = 0$, T fixed), the gel reaches mechanical equilibrium when the net driving force vanishes:

$$g(\phi, T) \equiv \Pi_{\rm el}(\phi) - \pi_{\rm osm}(\phi, T) = 0. \tag{10}$$

The function $g(\phi, T)$ can admit one or three roots depending on T, corresponding to monostable and bistable regimes respectively [Fig. 2(a,b)]. An equilibrium branch is mechanically stable when $\partial g/\partial\phi < 0$ (a restoring condition) and unstable when $\partial g/\partial\phi > 0$. The middle branch of the S-shaped equilibrium manifold is always unstable, giving rise to a hysteresis loop as T is quasi-statically varied.

Bistability requires the existence of a pair of fold (saddle-node) points, defined by $g = 0$ and $\partial g/\partial\phi = 0$ simultaneously. Eliminating ϕ from these two conditions yields a semi-analytic curve in parameter space [Fig. 2(c)] separating monostable from bistable regimes; closed-form expressions for $\partial g/\partial\phi$ are given in Appendix A.

Denoting the right-hand sides of Eqs. (5)–(8) as $F(\phi, T) \equiv (1/\tau_\phi) g(\phi, T)$ and $G(\phi, T) \equiv \gamma f(\phi) - k(T - T_0)$, the coupled dynamics admits fixed points (ϕ^*, T^*) satisfying both nullcline conditions:

$$g(\phi^*, T^*) = 0, \qquad T^* = T_0 + \frac{\gamma}{k} f(\phi^*). \tag{11}$$

Geometrically, fixed points are intersections of the swelling-equilibrium manifold $\mathcal{M}_g$ and the thermal nullcline $\mathcal{N}_T$ in the (ϕ, T) plane.

With the baseline parameters (Table I), the swelling relaxation time $\tau_\phi = 0.1\,$s is far shorter than the thermal relaxation time $1/k \approx 34\,$s, giving $\tau_\phi k \approx 3 \times 10^{-3} \ll 1$. This timescale separation is the geometric prerequisite for relaxation oscillations, and it is made explicit by introducing the small parameter $\varepsilon \equiv \tau_\phi k$ and the slow time $s \equiv kt$. In these variables the system takes the standard fast–slow form [25, 26]

$$\varepsilon \frac{d\phi}{ds} = g(\phi, T), \qquad \frac{dT}{ds} = \frac{\gamma}{k} f(\phi) - (T - T_0), \tag{12}$$

where $g(\phi, T) \equiv \Pi_{\rm el}(\phi) - \pi_{\rm osm}(\phi, T)$ is the swelling driving force as defined in Eq. (10).

In the singular limit $\varepsilon \to 0$, the dynamics decomposes into two sub-problems [25]. The *layer problem* (fast subsystem, T frozen) shows that ϕ relaxes instantaneously onto the critical manifold $\mathcal{M}_g$: $g(\phi, T) = 0$; the branch is attracting where $g_\phi < 0$ and repelling where $g_\phi > 0$. The *reduced problem* (slow sub-system, on the attracting branches of $\mathcal{M}_g$) uses the implicit relation $\phi = \phi_s(T)$ to eliminate ϕ and yields the one-dimensional slow flow

$$\frac{dT}{ds} = \frac{\gamma}{k} f(\phi_s(T)) - (T - T_0). \tag{13}$$

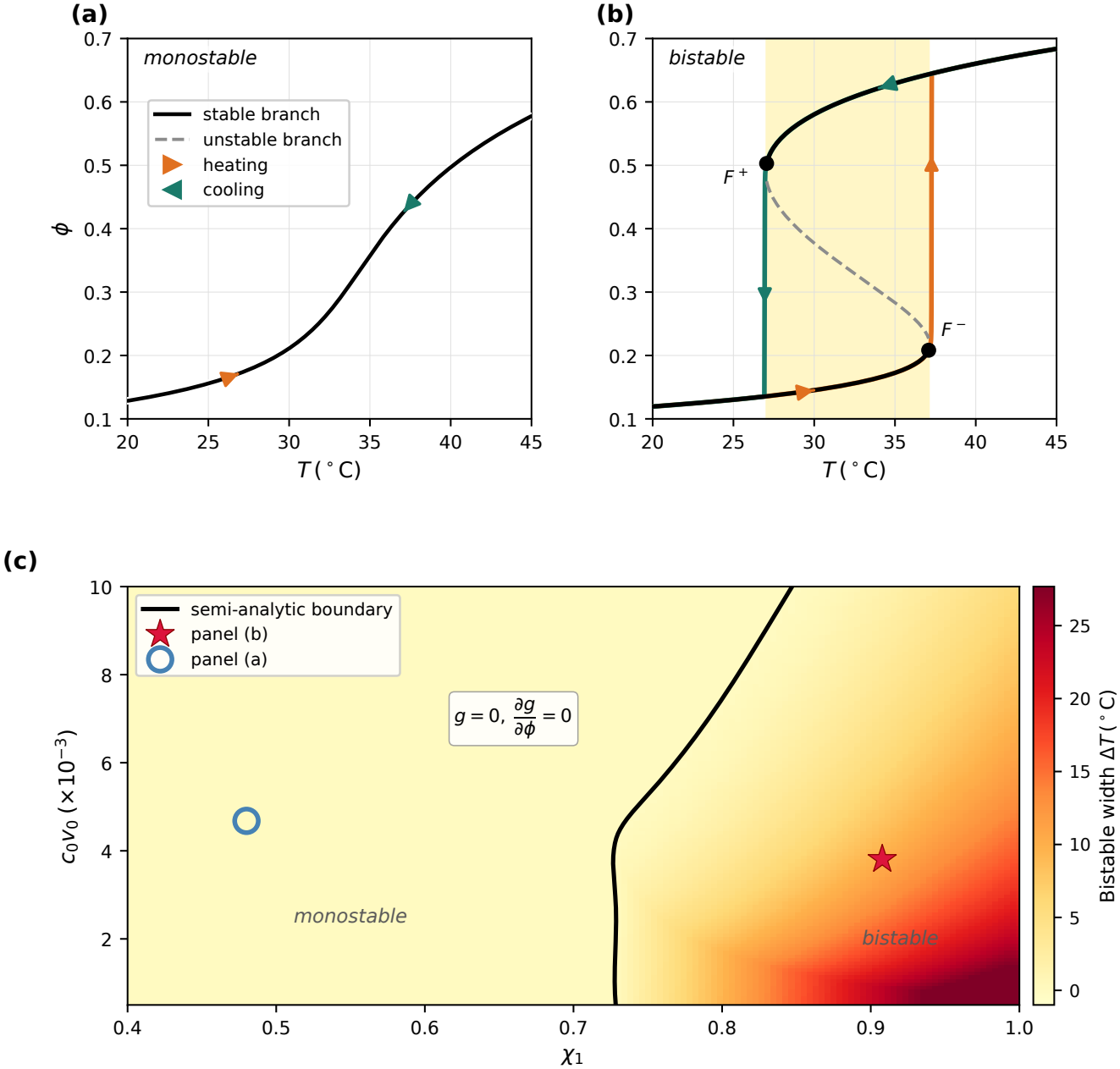

FIG. 2. Equilibrium swelling manifold and bistability phase diagram. (a) Monostable regime ($\chi_1 = 0.48$, $c_0 v_0 = 4.68 \times 10^{-3}$): a single physical root exists over the scanned temperature range; heating (orange) and cooling (teal) trajectories nearly coincide with no hysteresis. (b) Bistable regime ($\chi_1 = 0.908$, $c_0 v_0 = 3.81 \times 10^{-3}$): two stable branches (solid) and one unstable branch (dashed) coexist over a finite temperature interval bounded by fold points F^+ and F^-; quasi-static temperature sweeps produce a hysteresis loop with rapid jumps at the fold bifurcations. (c) Phase diagram in the (χ_1, $c_0 v_0$) plane. Color encodes the bistable temperature width ΔT; the semi-analytic boundary (black curve), obtained from the fold condition $g = 0$, $\partial g/\partial \phi = 0$, separates monostable (white) from bistable regions. Circle and star mark the parameter sets used in panels (a) and (b), respectively.

The fold points of $\mathcal{M}_g$, where $g = 0$ and $g_\phi = 0$ simultaneously, are the switching points at which the slow flow terminates and a rapid $O(\varepsilon)$-time jump to the opposite attracting branch occurs. The relaxation limit cycle is therefore assembled from two slow-flow segments (on the swollen and collapsed branches of $\mathcal{M}_g$) connected by two near-vertical fast jumps at the fold points. Because the cycle is a large-amplitude global object controlled by the fold-point temperatures and the slow-flow accumulation time $\sim 1/k$, its existence boundary (the LPC branch $\mathcal{L}$) lies far from the Hopf point and cannot be predicted from local normal-form analysis alone.

B. Linear stability and Hopf bifurcation

To obtain the local stability criterion, we linearize F and G around (ϕ^*, T^*) with perturbations $(\delta\phi, \delta T)^T$. The variational system is

$$\frac{d}{dt}\begin{pmatrix}\delta\phi\\ \delta T\end{pmatrix} = J_*\begin{pmatrix}\delta\phi\\ \delta T\end{pmatrix}, \qquad J_* = \begin{pmatrix}F_\phi & F_T\\ G_\phi & G_T\end{pmatrix}_*, \tag{14}$$

with

$$F_\phi = \frac{1}{\tau_\phi}\left[\Pi'_{\rm el}(\phi) - \pi_\phi(\phi, T)\right], \quad F_T = -\frac{1}{\tau_\phi}\pi_T(\phi, T),$$
$$G_\phi = \gamma f'(\phi), \qquad G_T = -k, \tag{15}$$

where $\pi_\phi = \partial\pi_{\rm osm}/\partial\phi$ and $\pi_T = \partial\pi_{\rm osm}/\partial T$. The characteristic polynomial is

$$\lambda^2 - ({\rm tr}J_*)\lambda + \det J_* = 0, \tag{16}$$

with

$${\rm tr}J_* = \frac{1}{\tau_\phi}\left[\Pi'_{\rm el}(\phi^*) - \pi_\phi(\phi^*, T^*)\right] - k, \tag{17}$$

$$\det J_* = -\frac{k}{\tau_\phi}\left[\Pi'_{\rm el}(\phi^*) - \pi_\phi(\phi^*, T^*)\right] + \frac{\gamma}{\tau_\phi} f'(\phi^*)\pi_T(\phi^*, T^*). \tag{18}$$

Hence the standard two-dimensional criteria apply [26]: stable fixed point if ${\rm tr}J_* < 0$ and $\det J_* > 0$; saddle if $\det J_* < 0$; Hopf onset when

$${\rm tr}J_* = 0, \qquad \det J_* > 0, \tag{19}$$

with Hopf frequency $\omega_H = \sqrt{\det J_*}$.

The Hopf condition ${\rm tr}J_* = 0$, together with Eq. (17), implies

$$g_\phi(\phi^*, T^*) \equiv \Pi'_{\rm el}(\phi^*) - \pi_\phi(\phi^*, T^*) = k\tau_\phi > 0, \tag{20}$$

so the Hopf fixed point necessarily lies on the *mechanically destabilizing* segment of the equilibrium manifold—where the isolated $\dot\phi$ subsystem alone would be unstable—but is stabilized by thermal damping until the feedback overcomes the cooling. A genuine oscillatory instability (rather than a saddle) additionally requires $\det J_* > 0$. Substituting Eq. (20) into Eq. (18) shows that $\det J_* > 0$ demands

$$\frac{\gamma}{\tau_\phi} f'(\phi^*)\,\pi_T(\phi^*, T^*) > k^2. \tag{21}$$

The thermal derivative evaluates to $\pi_T = \Delta h\,\phi^2/(k_B T^2)$ (see Appendix B 1). Because $\Delta h < 0$ for the LCST polymer (Table I), we have $\pi_T < 0$; and since $f'(\phi^*) < 0$ for $\phi^* > \phi_g$ (the Hill gate is decreasing), the product $f'\pi_T > 0$. Condition (21) can therefore be satisfied for sufficiently large heating rate γ, which explains why the Hopf curve $\mathcal{H}$ exists only above a threshold in γ (Fig. 4). Below this threshold the fixed point on the destabilizing branch is a saddle, not a spiral, and no Hopf instability occurs.

To determine the Hopf criticality, we reduce the dynamics near Eq. (19) to the center manifold [26–28]. The amplitude equation takes the normal-form expansion

$$\dot r = r\left(\mu + l_1 r^2 + l_2 r^4 + O(r^6)\right), \qquad \mu = \frac{1}{2}{\rm tr}J_*, \tag{22}$$

where l_1 and l_2 are the first and second Lyapunov coefficients. The sign of l_1 determines the Hopf type [27]: $l_1 < 0$ supercritical, $l_1 > 0$ subcritical.

Numerical evaluation of l_1 along the entire Hopf branch $\mathcal{H}$ (Fig. 3; details in Appendix B) shows that l_1 is strictly positive at all computed points, confirming that the Hopf bifurcation is subcritical throughout. No Bautin (generalized Hopf, $l_1 = 0$) codimension-2 point exists in the explored window [27]. Therefore no stable small-amplitude cycle is born at onset; the attracting relaxation cycle is selected by global periodic-orbit geometry (LPC/BP branches), not by local Hopf onset alone.

The one-dimensional bifurcation diagram along $k = 0.0295$ (Fig. 3) illustrates the consequences of subcriticality. The stable oscillation region is bounded not by the Hopf curve $\mathcal{H}$ but by the Limit Point of Cycles (LPC) branch at $\gamma_{\rm LPC} \approx 2.94$, giving, along $k = 0.0295\,{\rm s}^{-1}$, a bistable window $\Delta\gamma \approx 4.5$ between the LPC and Hopf points.

C. Periodic-orbit boundaries, phase diagram, and phase-plane structure

Local bifurcations of equilibria (fold and Hopf) do not, by themselves, delimit the region where stable periodic orbits exist. We therefore track periodic-orbit branches via numerical continuation (AUTO-07p) [29] and identify three key boundaries in the (γ, k) parameter plane:

1. $\mathcal{H}$: Hopf branch of fixed points (${\rm tr}\,J = 0$, $\det J > 0$).

2. $\mathcal{L}$: Limit Point of Cycles (LPC), where stable and unstable periodic orbits collide and annihilate [27, 29].

3. $\mathcal{B}$: Branch Point (BP) connected periodic branch, a secondary family of periodic orbits meeting $\mathcal{L}$ at the codimension-2 organizing point.

The oscillation boundary is defined by the outer envelope of $\mathcal{L}$ and $\mathcal{B}$, rather than by $\mathcal{H}$ directly. Numerically, $\mathcal{L}$ and $\mathcal{B}$ both terminate at the codimension-2 point $(\gamma, k) \approx (1.24, 0.0139)$, which organizes the oscillation boundary at small parameter values (Fig. 4).

Using the bifurcation boundaries established above, we classify the (γ, k) plane into three dynamical regimes: Fixed-Point only (FP-only), Limit-Cycle only (LC-only), and Bistable, as shown by the shaded regions in Fig. 4.

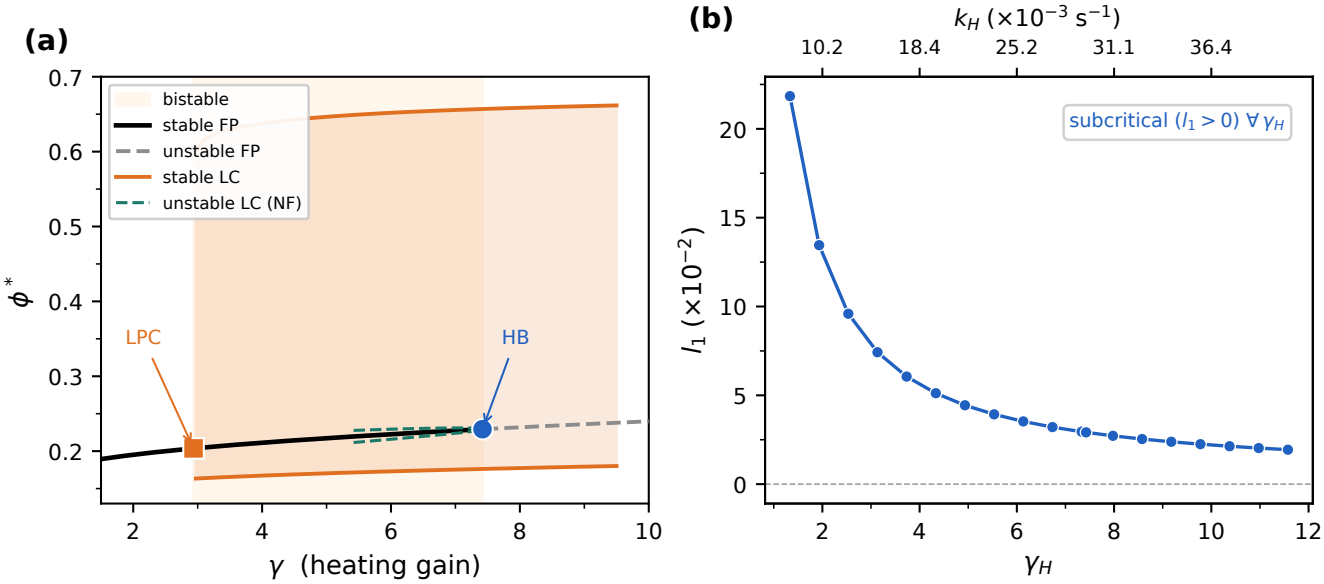

FIG. 3. Subcritical bifurcation structure and Lyapunov coefficient. (a) One-dimensional bifurcation diagram at $k = 0.0295\,\mathrm{s}^{-1}$. Stable (black solid) and unstable (gray dashed) fixed-point branches, stable limit-cycle envelope $\phi_{\max}, \phi_{\min}$ (orange), and unstable limit-cycle branch (teal dashed, normal-form near HB). The subcritical Hopf bifurcation (HB) at $\gamma_H = 7.42$ and the limit point of cycles (LPC) at $\gamma_{\mathrm{LPC}} = 2.94$ delimit the bistable region (shaded). (b) First Lyapunov coefficient l_1 along the Hopf branch $\mathcal{H}$; $l_1 > 0$ at all computed points confirming subcriticality throughout. Top axis: corresponding cooling rate k_H.

The classification is validated by two long-time integrations per grid point—one from a default initial condition and one from a fixed-point-near condition—to distinguish true coexistence (bistable) from LC-only behavior (see Appendix C).

Within the oscillatory region, the period increases monotonically with γ and varies non-monotonically with k (Fig. 3). The stable limit cycle exists only inside the envelope defined by $\mathcal{L}$ (LPC) and $\mathcal{B}$; the Hopf curve $\mathcal{H}$ marks the fixed-point stability boundary and lies *inside* this envelope. On the high-γ side, as $\gamma \to \gamma_{\mathcal{L}}^{+}$ from within the LC-only region (approaching the LPC boundary), the stable and unstable limit cycles coalesce and annihilate: this fold of cycles carries no period divergence, so the period there remains finite. On the low-γ side the period is also finite and controlled primarily by the thermal accumulation time $1/k$—the dominant slow-variable timescale in the relaxation cycle. The non-monotonic dependence on k reflects the competing effects of faster cooling on the thermal accumulation time and on the LPC boundary location. Quantitative period and amplitude trends are shown in Fig. 3(b,c).

To interpret the three regimes physically, we visualize the phase flow in the (ϕ, T) plane. The equilibrium swelling manifold $\mathcal{M}_g$ ($g = 0$) and the thermal nullcline $\mathcal{N}_T$ ($\dot{T} = 0$) partition the plane, with their intersections defining the fixed points.

In the FP-only regime [Fig. 4(b)], all trajectories collapse to the stable fixed point on $\mathcal{M}_g$. In the LC-only regime [Fig. 4(c)], a stable relaxation oscillation emerges through slow drift along stable manifold branches punctuated by rapid jumps at fold points—a classic relaxation cycle; the fixed point is unstable. In the bistable regime [Fig. 4(d)], a stable fixed point and a stable limit cycle coexist, with initial conditions determining which attractor is reached. This phase-plane picture provides a transparent geometric interpretation of the bifurcation structures established in the preceding subsections.

IV. RESULTS AND DISCUSSION

A. Physical mechanism of oscillations

The bifurcation structure established in Sec. III determines *where* stable oscillations exist; we now examine *what* those oscillations look like and why they take the form they do. We fix representative parameters $\gamma = 3.497\,^{\circ}\mathrm{C\,s}^{-1}$ and $k = 0.0275\,\mathrm{s}^{-1}$ (LC-only region, $T^* \approx 31.3\,^{\circ}\mathrm{C}$) and dissect one complete limit cycle (period $P = 125.4\,\mathrm{s}$, $\phi \in [0.166, 0.637]$, $T \in [26.1, 41.9]\,^{\circ}\mathrm{C}$) into four dynamically distinct stages using the phase plane, time series, and energy-flow decomposition shown in Fig. 5.

Stage I: slow swollen drift (12.5% of P, 15.7 s).— The state resides on the swollen stable branch of $\mathcal{M}_g$ at low ϕ, where the gate function $f \approx 1$ and internal heating $\gamma f(\phi)$ slightly exceeds environmental heat loss $k(T - T_0)$. Temperature rises slowly along this branch [Fig. 5(a,b)] until it approaches the upper fold F^+ ($T \approx 37.1\,^{\circ}\mathrm{C}$),

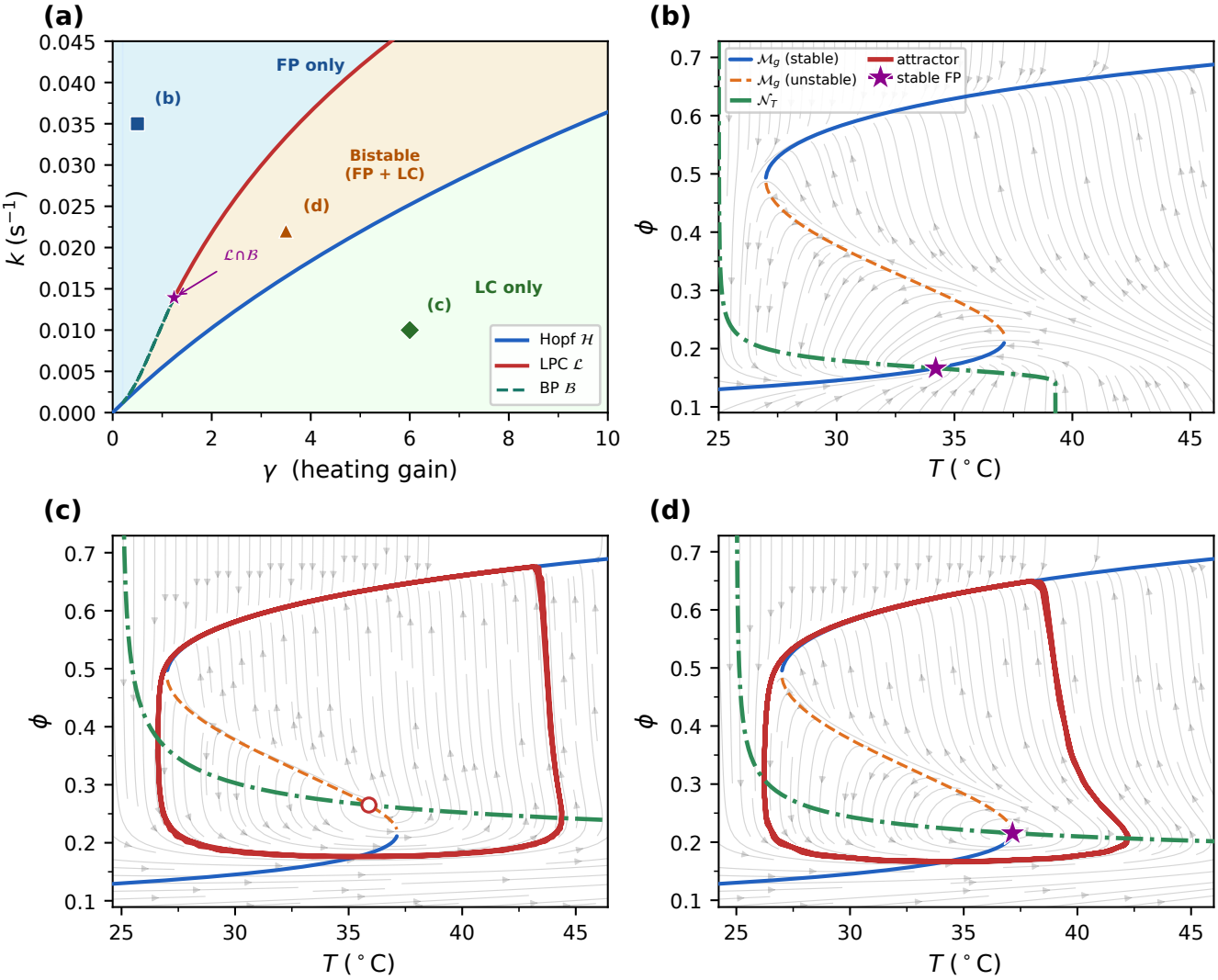

FIG. 4. Phase diagram and representative phase portraits. (a) Phase diagram in the (γ, k) plane. The Hopf branch $\mathcal{H}$ (blue), LPC branch $\mathcal{L}$ (red), and BP branch $\mathcal{B}$ (green dashed) delineate three dynamical regimes: FP only, LC only, and Bistable. $\mathcal{L}$ and $\mathcal{B}$ terminate at the codimension-2 point $(\gamma, k) \approx (1.24, 0.014)$. Markers indicate the parameter values used in panels (b)–(d). (b) FP-only regime at $(\gamma, k) = (0.33,\ 0.0062\,\mathrm{s}^{-1})$: all trajectories collapse to the stable fixed point ($\star$) on $\mathcal{M}_g$. (c) LC-only regime at $(\gamma, k) = (5.84,\ 0.0182\,\mathrm{s}^{-1})$: the attractor is a relaxation cycle (red loop) with slow drift along stable branches of $\mathcal{M}_g$ and rapid jumps at fold points; the fixed point ($\circ$) is unstable. (d) Bistable regime at $(\gamma, k) = (5.10,\ 0.0252\,\mathrm{s}^{-1})$: a stable fixed point ($\star$) and a stable limit cycle (red loop) coexist. Faint trajectories in (c) and (d) are seeded from random initial conditions and colored by their asymptotic attractor (blue $\to$ FP, orange $\to$ LC).

beyond which no swollen equilibrium exists. In the singular limit $\varepsilon \equiv \tau_\phi k \to 0$ ($\varepsilon \approx 0.003$ here), the jump occurs exactly at F^+; at finite ε it is triggered in the immediate vicinity of F^+, so the cycle's maximum temperature ($T_{\max} \approx 41.9\,°\mathrm{C}$) exceeds $T(F^+)$ by the thermal inertia accumulated during the finite-duration collapse (Stage II).

Stage II: fast collapse (8.6% of P, 10.8 s).— Loss of the swollen branch at F^+ forces a rapid jump to the collapsed branch. Because swelling dynamics are governed by the overdamped equation (5) with timescale $\tau_\phi = 0.1\,\mathrm{s}$, ϕ rises from ≈ 0.17 to ≈ 0.64 on a timescale much faster than the thermal relaxation; T changes negligibly during this jump [Fig. 5(a)].

Stage III: slow collapsed drift (64.8% of P, 81.3 s).— On the collapsed branch, $f(\phi) \approx 0$ and internal heating is suppressed. Heat loss $k(T - T_0)$ now drives T downward exponentially with time constant $1/k \approx 36\,\mathrm{s}$, from the post-collapse temperature ($T \approx 41.9\,°\mathrm{C}$) toward the lower fold F^- ($T \approx 27.0\,°\mathrm{C}$), a descent of $\approx 15\,°\mathrm{C}$ [Fig. 5(a,b)]. This stage dominates the period because the gel must passively cool across the full inter-fold temperature interval; there is no active heat input to hasten the transition.

Stage IV: fast re-swelling (14.1% of P, 17.6 s).— At F^- the collapsed branch disappears and the state jumps back to the swollen branch, completing the cycle. The re-swelling jump is somewhat slower than the collapse because ϕ must decrease by a factor of ≈ 4, traversing a wider range on the manifold.

The pronounced asymmetry—Stage III alone accounts for 65% of the period—has a direct physical origin: during collapse the gate is closed and cooling is purely passive, whereas during Stage I the gate is open and active heating drives the system toward F^+ on a timescale set by $(\gamma/k)(f\,/\,\Delta T)$, which is considerably shorter than $1/k$. Consequently, the swollen-state residence time is determined by the heating rate γ, while the collapsed-state residence time is controlled by the cooling rate k and the inter-fold temperature gap $\Delta T_F \equiv F^+ - F^- \approx 10.1\,^\circ\mathrm{C}$. As k increases, Stage III shortens proportionally, redistributing time toward the fast jump stages; the stacked-fraction diagram [Fig. 5(d)] confirms that the Stage III fraction decreases from $\approx 78\%$ at $k = 0.022\,\mathrm{s}^{-1}$ to $\approx 58\%$ at $k = 0.040\,\mathrm{s}^{-1}$, with Stages II and IV growing in concert.

The energy-flow decomposition [Fig. 5(c)] provides an independent self-consistency check. The instantaneous heating $Q_{\mathrm{in}} = \gamma f(\phi)$ is concentrated in Stages I and IV (gate open), while $Q_{\mathrm{out}} = k(T - T_0)$ is sustained throughout Stage III (elevated T). Over one period, numerical integration gives $\oint Q_{\mathrm{in}}\,dt = \oint Q_{\mathrm{out}}\,dt = 21.7\,^\circ\mathrm{C}$ (balance error $< 0.2\%$), confirming that the limit cycle is energetically self-consistent. This balance is a necessary consequence of the periodicity of $T(t)$, not an independent constraint on the oscillation domain; the latter is set by the global bifurcation boundaries $\mathcal{L}$ and $\mathcal{B}$ identified in Sec. III C.

B. Parameter control of oscillation characteristics

The preceding section established the four-stage anatomy of a representative relaxation cycle. We now ask how the quantitative characteristics of the oscillation—period, amplitude, and waveform—respond to the three key parameters (γ, k, n). The central result is that these three observables are controlled by largely independent physical mechanisms, providing decoupled experimental handles.

Period: set by k, insensitive to γ.— In the singular limit $\varepsilon \to 0$, the period is dominated by Stage III, in which the state drifts passively along the collapsed branch under pure environmental cooling. Integrating $\dot{T} = -k(T - T_0)$ from F^+ to F^- gives the leading-order period estimate

$$P_0 = \frac{1}{k} \ln \frac{F^+ - T_0}{F^- - T_0} \approx \frac{1}{k} \ln \frac{37.1 - T_0}{27.0 - T_0}, \tag{23}$$

where the logarithmic prefactor is a geometric constant of $\mathcal{M}_g$ independent of γ and k. Equation (23) predicts $P \propto k^{-1}$. Fitting the period map (Fig. 6a) over the full oscillatory region yields $P \sim k^{-0.84}$, in approximate agreement; the sub-unity exponent reflects the finite contribution of Stages I and IV ($\approx$ 20–27% of P), which carry their own k-dependent time constants. By contrast, varying γ at fixed k gives $P \sim \gamma^{0.02}$ (Fig. 6a, near-horizontal iso-period contours), confirming that heating rate has negligible influence on the period. The physical reason is gate saturation: during Stage I the gate function $f \approx 1$, so increasing γ proportionally accelerates Stage I, but this stage represents only $\approx 13\%$ of the total period; the dominant Stage III is entirely independent of γ because $f \approx 0$ throughout.

Amplitude: fixed by manifold geometry, robust to parameter variation.— The oscillation amplitude $\Delta\phi \equiv \phi_{\max} - \phi_{\min}$ is determined to leading order by the inter-fold span of the stable branches of $\mathcal{M}_g$: the cycle must traverse from the swollen branch near $\phi(F^+) \approx 0.21$ to the collapsed branch near $\phi(F^-) \approx 0.50$. Accordingly, $\Delta\phi$ varies only weakly across the entire oscillatory domain: scanning the period–amplitude map (Fig. 6b) gives $\Delta\phi \in [0.45, 0.49]$ with a coefficient of variation of $\approx 2\%$, irrespective of γ or k. This near-constancy has an important implication at the oscillation boundary. As γ decreases toward the LPC boundary $\mathcal{L}$, $\Delta\phi$ remains finite ($\approx$ 0.45–0.48 across all k slices) rather than decreasing continuously to zero. The finite-amplitude disappearance of the oscillation at $\mathcal{L}$ is the direct signature of a subcritical LPC: the stable and unstable limit cycles annihilate in a saddle-node bifurcation of cycles, carrying no amplitude precursor. In contrast, a supercritical Hopf bifurcation would produce small-amplitude cycles that grow continuously from zero; the present model predicts that such small-amplitude oscillations do not exist.

Waveform: shaped by Hill exponent n.— While k and $\mathcal{M}_g$ geometry control the timescale and amplitude, the Hill exponent n of the gate function (9) governs the waveform asymmetry and the extent of the oscillatory domain. A larger n sharpens the collapse–swelling transition, which accelerates the activation and deactivation of internal heating. Two consequences follow directly from Fig. 6(c,d). First, the lower LPC boundary shifts upward with n: $\gamma_{\mathrm{LPC}} \approx 0.56\,n + 0.08$, meaning a sharper gate requires a stronger heating drive to sustain oscillations. Second, the Stage III fraction decreases monotonically from 86% at $n = 2$ to 57% at $n = 12$ (Fig. 6d), while Stages II and IV grow symmetrically. Physically, larger n causes the gate to close more abruptly at collapse, allowing the collapsed-state temperature to recover more quickly before re-swelling; the waveform therefore becomes less asymmetric as n increases, with the period shortening from 261 s ($n = 2$) to 124 s ($n = 12$) at fixed $\gamma = 8.0\,^\circ\mathrm{C\,s}^{-1}$. Experimentally, n corresponds to the cooperativity of the LCST transition or the nonlinearity of the photothermal conversion process, and is therefore a natural target for materials-level tuning of the oscillation waveform.

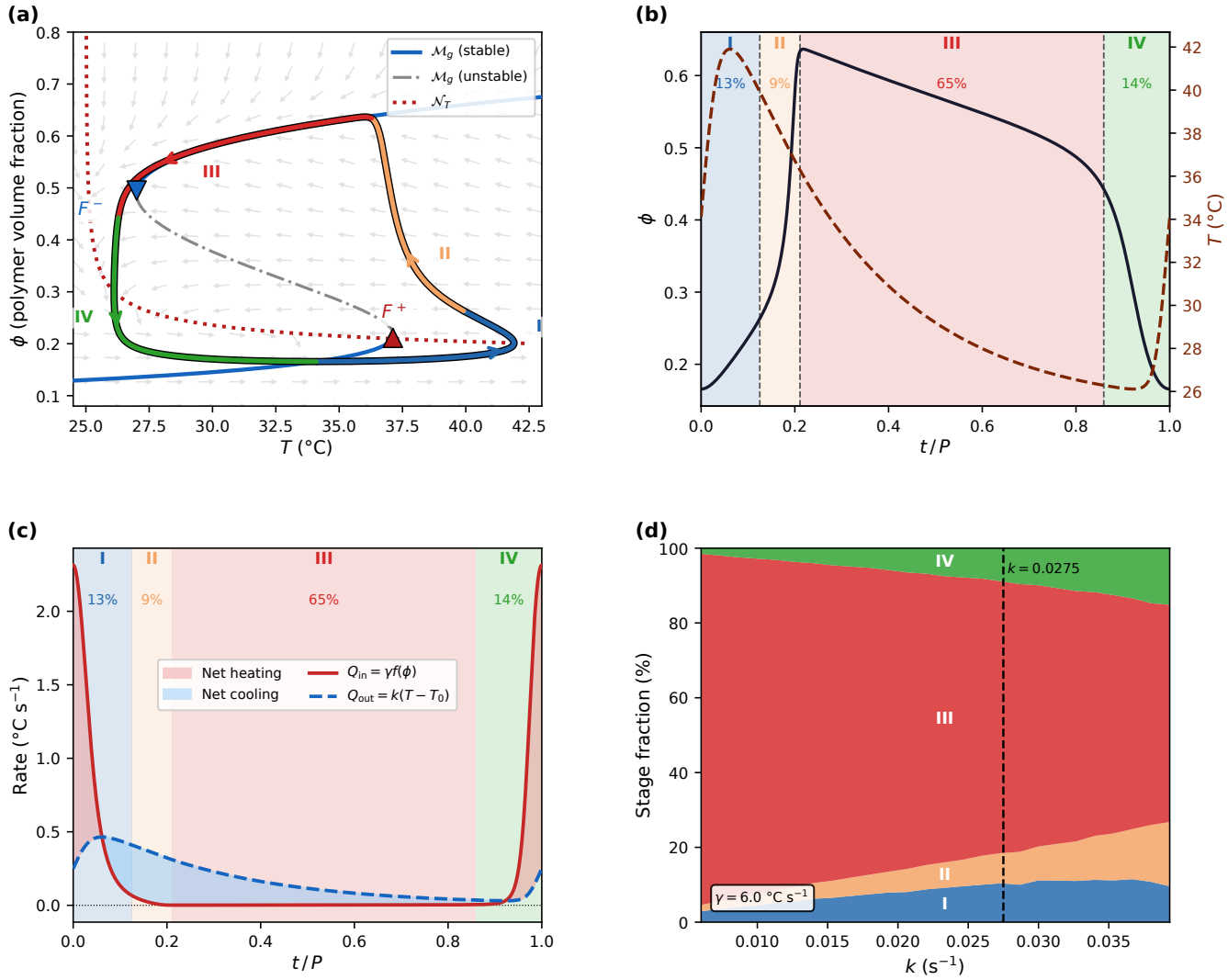

FIG. 5. Anatomy of the relaxation oscillation cycle at $\gamma = 3.497\,^\circ\mathrm{C\,s^{-1}}$, $k = 0.0275\,\mathrm{s^{-1}}$ ($P = 125.4\,\mathrm{s}$). (a) Phase plane (T–ϕ). The equilibrium swelling manifold $\mathcal{M}_g$ ($g = 0$; blue solid for both stable branches, gray dash-dot for the unstable middle branch) and the thermal nullcline $\mathcal{N}_T$ (red dotted) are shown together with the limit cycle, colour-coded by dynamical stage: Stage I (blue, swollen slow), II (orange, fast collapse), III (red, collapsed slow), IV (green, fast re-swell). Arrows indicate the direction of traversal. The upper fold point F^+ (△, red) and lower fold point F^- (▽, blue) mark the ends of the unstable branch of $\mathcal{M}_g$; in the singular limit $\tau_\phi/\tau_T \to 0$ the fast jumps occur precisely at these points, while at finite $\varepsilon = \tau_\phi k \approx 0.003$ the actual jump is triggered in their vicinity. The background quiver shows the normalized phase-plane flow. (b) Time series of $\phi(t)$ (dark, left axis) and $T(t)$ (brown dashed, right axis) over one normalised period t/P. Stage boundaries (vertical dashed lines) and fractional durations are indicated. (c) Instantaneous energy-flow rates $Q_{\mathrm{in}} = \gamma f(\phi)$ (red solid) and $Q_{\mathrm{out}} = k(T - T_0)$ (blue dashed); shaded areas mark net heating (pink) and net cooling (blue) intervals. Stage boundaries are indicated as in (b). (d) Stage fractions as a function of cooling rate k at fixed $\gamma = 6.0\,^\circ\mathrm{C\,s^{-1}}$. The collapsed-slow stage III (red) dominates at low k and shrinks monotonically as k increases. The reference value $k = 0.0275\,\mathrm{s^{-1}}$ used in panels (a)–(c) is marked.

C. Robustness to gate function form

The results in Secs. IV A and IV B were derived for a Hill-type gate function with exponent $n = 3$ and steepness $\rho = 30$. We now ask whether the existence of the oscillatory region and the topology of its boundaries are specific to this choice, or whether they persist for qualitatively different functional forms. Three gate functions were compared (Fig. 7b): the Hill function $f_{\mathrm{H}}(\phi) = (1 + (\rho(\phi - \phi_g))^n)^{-1}$; a hyperbolic tangent $f_{\mathrm{T}}(\phi) = \frac{1}{2}(1 - \tanh(s(\phi - \phi_{1/2})))$; and a piecewise-linear ramp $f_{\mathrm{P}}(\phi)$. The three steepness parameters ($n = 3$, $\rho = 30$; $s = 41.5$; slope $\Delta\phi_{10\%\to90\%} = 0.053$) were chosen to yield equal transition widths $\Delta\phi_{10\%\to90\%} = 0.053$, isolating the effect of functional form from that of transition width.

LPC boundary: topologically robust, quantitatively

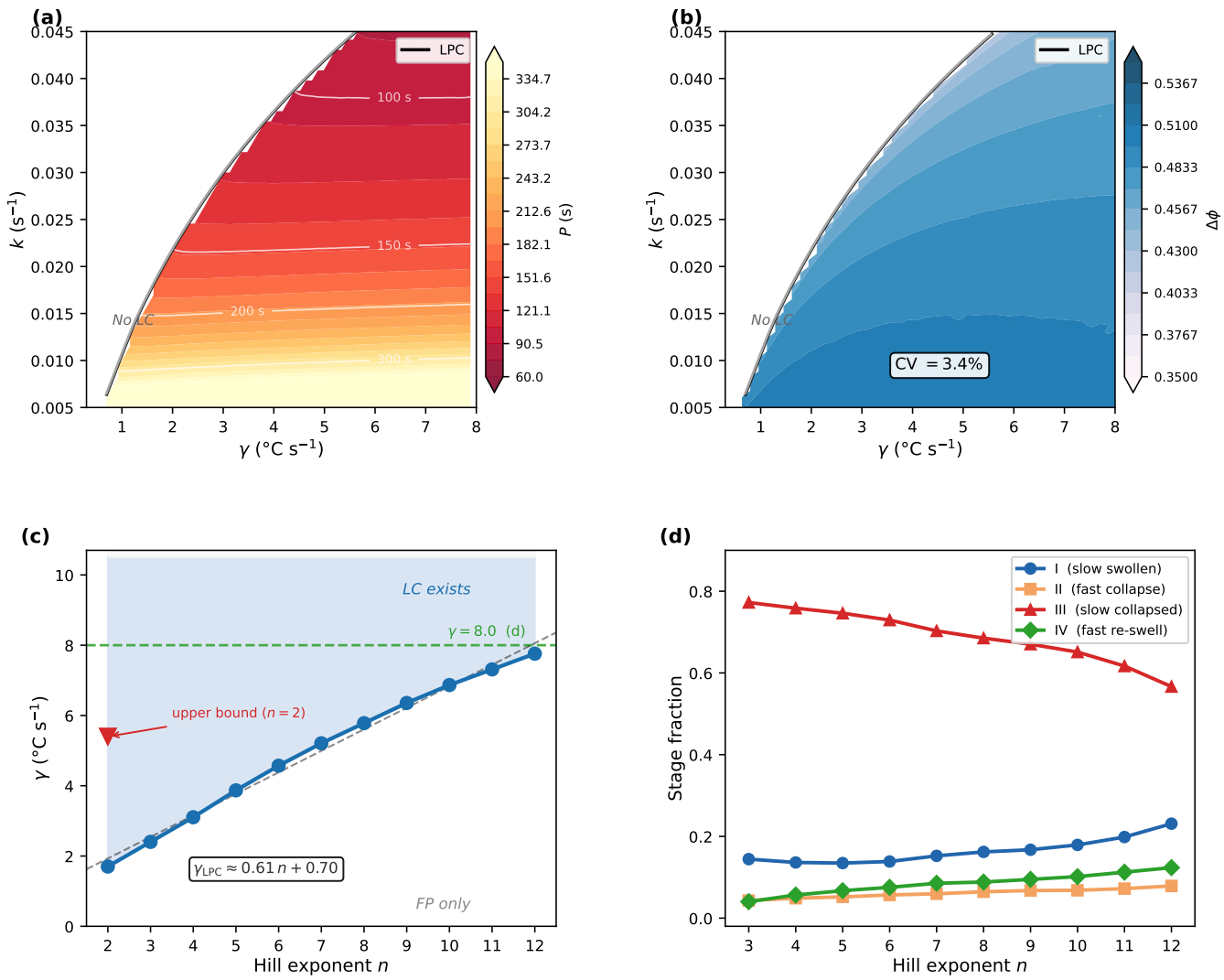

FIG. 6. Parameter control of oscillation period, amplitude, and waveform. (a) Period P in the (γ, k) plane. The oscillatory region (right of the LPC boundary, black curve) is colour-coded by P (s). White iso-period contours are labelled in seconds. The near-horizontal alignment of contours demonstrates that P is controlled primarily by k ($P \sim k^{-0.84}$) and is insensitive to γ ($P \sim \gamma^{0.02}$). (b) Oscillation amplitude $\Delta\phi \equiv \phi_{\max} - \phi_{\min}$ in the same (γ, k) plane. The coefficient of variation across the oscillatory domain is $\approx 2\%$, confirming that amplitude is set by the inter-fold geometry of $\mathcal{M}_g$ and is robust to parameter variation. The finite value of $\Delta\phi$ at the LPC boundary (left edge of coloured region) is the hallmark of subcritical LPC: oscillations disappear with finite amplitude, not continuously. (c) LPC threshold $\gamma_{\rm LPC}$ as a function of Hill exponent n (filled circles, left axis). The blue shaded band is the oscillatory region ($\gamma_{\rm LPC} < \gamma < \gamma_{\max}$). A linear fit gives $\gamma_{\rm LPC} \approx 0.56\,n + 0.08$ (dashed). The horizontal dashed line marks $\gamma = 8.0\,^\circ\mathrm{C\,s^{-1}}$, the value used in panel (d). (d) Stage fractions as a function of n at $\gamma = 8.0\,^\circ\mathrm{C\,s^{-1}}$, $k = 0.0275\,\mathrm{s^{-1}}$. The Stage III (collapsed slow, red) fraction decreases from 86% ($n = 2$) to 57% ($n = 12$), quantifying the waveform symmetrisation with increasing gate sharpness. Stages II and IV grow in concert.

shifted.— For all three gate functions, a LPC boundary exists across the full range of k examined (Fig. 7a). The boundary shape—a monotonically increasing curve in the (γ, k) plane—is preserved in all three cases. The quantitative position shifts systematically with gate smoothness: at $k = 0.025\,\mathrm{s^{-1}}$, $\gamma_{\rm LPC} = 2.37$, 3.06, and $3.74\,^\circ\mathrm{C\,s^{-1}}$ for the Hill, tanh, and piecewise-linear forms, respectively, giving a total spread of $\Delta\gamma_{\rm LPC} = 1.37\,^\circ\mathrm{C\,s^{-1}}$. The physical origin of this trend is the local feedback gain $|\partial f/\partial\phi|$ in the transition region: the smoother the gate, the lower the peak gain, and the larger the heating amplitude γ required to sustain the oscillation against environmental cooling. Crucially, however, the oscillatory region does not disappear for any of the three forms; it merely shifts.

Hopf boundary: a gate-smoothness–dependent gradient from subcritical to globally driven.— While the LPC boundaries cluster within $\Delta\gamma \approx 1.4\,^\circ\mathrm{C\,s^{-1}}$, the Hopf boundaries diverge dramatically across the three gate forms. For the Hill function, the Hopf boundary lies at

$\gamma_{\mathrm{H}} \approx 5$–$8\,^{\circ}\mathrm{C\,s^{-1}}$ across the oscillatory k range and is visible in Fig. 7a as the thin blue curve inside the LPC boundary: the fixed point loses stability before the LPC appears, the canonical subcritical scenario. For the tanh gate, γ_{H} increases steeply with k, reaching $\approx 30\,^{\circ}\mathrm{C\,s^{-1}}$ at $k = 0.025\,\mathrm{s^{-1}}$ and $\approx 80\,^{\circ}\mathrm{C\,s^{-1}}$ at $k = 0.04\,\mathrm{s^{-1}}$, far outside the plot range; only the lowest-k portion of the Hopf curve is visible (thin red segment). For the piecewise-linear gate, no Hopf bifurcation was found for $\gamma < 120\,^{\circ}\mathrm{C\,s^{-1}}$: the fixed point remains linearly stable throughout the entire oscillatory region.

This progression has a direct physical interpretation. As the gate becomes less smooth, the Jacobian element $\partial^2 F/\partial\phi^2|_{\phi=\phi^*}$ that drives the Hopf instability decreases; the onset of linear instability is displaced to ever-larger γ. In the piecewise-linear limit, the gate is linear on the transition interval and the relevant Jacobian contribution vanishes entirely, so no Hopf bifurcation occurs. The oscillation in this case originates purely from the global geometry of $\mathcal{M}_g$: the slow manifold folds at $F^{\pm}$ force the trajectory to jump between branches, generating a relaxation cycle without any local instability of the fixed point. This demonstrates that the *necessary condition* for self-oscillation is the global topology—the existence of a folded slow manifold and sufficient heating gain $\gamma > \gamma_{\mathrm{LPC}}$—rather than local Hopf instability, which is a gate-specific feature that may or may not accompany the LPC.

V. CONCLUSION

We have analyzed the self-oscillation of a minimal two-variable thermo-swelling gel model in which a collapse-suppressed heating gate couples to overdamped swelling kinetics and environmental cooling. The analysis proceeds in three layers: the equilibrium geometry of the swelling manifold, the bifurcation structure of its fixed points and periodic orbits, and the physical anatomy and parametric sensitivity of the resulting relaxation cycle.

The equilibrium manifold $\mathcal{M}_g$ is S-shaped in the (ϕ, T) plane, with two fold points $F^{\pm}$ that demarcate stable and unstable branches. Fixed-point stability analysis shows that the unique fixed point can undergo a Hopf bifurcation as the heating rate γ is increased. Normal-form computation of the first Lyapunov coefficient $L_1 > 0$ establishes that the Hopf bifurcation is subcritical: no small-amplitude limit cycle bifurcates from the fixed point; instead, the fixed point loses stability in the presence of a pre-existing finite-amplitude limit cycle. The stable oscillatory domain is therefore bounded not by the Hopf curve but by a limit-point-of-cycles (LPC) boundary, at which the stable and unstable limit cycles annihilate in a saddle-node bifurcation of cycles. Between the LPC and Hopf boundaries, a stable limit cycle coexists with a stable fixed point, producing a genuine bistable region that has no counterpart in supercritical scenarios.

The relaxation cycle consists of four stages organized by the fast–slow geometry of $\mathcal{M}_g$. In the singular limit $\varepsilon = \tau_\phi k \to 0$, the slow stages (I and III, along the stable manifold branches) are separated by instantaneous jumps at F^+ and F^-. At representative parameters ($\gamma = 3.50\,^{\circ}\mathrm{C\,s^{-1}}$, $k = 0.0275\,\mathrm{s^{-1}}$), the period is $P = 125\,\mathrm{s}$, with Stage III (passive cooling along the collapsed branch) accounting for 64.8% of the cycle. The strong asymmetry is a direct consequence of the feedback gate: heating is active ($f \approx 1$) only during Stage I, whereas cooling is passive throughout Stage III ($f \approx 0$), giving a characteristic cooling timescale $1/k \approx 36\,\mathrm{s}$ that dominates the period.

Parameter sensitivity analysis separates three independently controllable characteristics. The period is determined primarily by k, with $P \sim k^{-0.84}$ (close to the singular-limit prediction $P \propto k^{-1}$), and is insensitive to γ ($P \sim \gamma^{0.02}$). The amplitude $\Delta\phi$ is fixed by the inter-fold geometry of $\mathcal{M}_g$ and varies by only $\approx 2\%$ across the entire oscillatory region; its finite value at the LPC boundary is the hallmark signature of the subcritical transition. The waveform symmetry is controlled by the Hill exponent n: increasing n from 2 to 12 reduces the Stage III fraction from 86% to 57% and shifts the LPC threshold as $\gamma_{\mathrm{LPC}} \approx 0.56\,n + 0.08$.

Comparison of three gate functions with equal transition width ($\Delta\phi_{10\%\to 90\%} = 0.053$) reveals that the oscillatory region is topologically robust: LPC boundaries exist for Hill, tanh, and piecewise-linear gates across the full parameter range, shifting by $\Delta\gamma_{\mathrm{LPC}} \leq 1.37\,^{\circ}\mathrm{C\,s^{-1}}$. The Hopf boundary, however, diverges dramatically with gate smoothness. For the Hill gate it lies within the diagram at $\gamma_{\mathrm{H}} \approx 5$–$8\,^{\circ}\mathrm{C\,s^{-1}}$; for the tanh gate it recedes to $\gamma_{\mathrm{H}} \gtrsim 30\,^{\circ}\mathrm{C\,s^{-1}}$; and for the piecewise-linear gate no Hopf bifurcation was found for $\gamma < 120\,^{\circ}\mathrm{C\,s^{-1}}$. This progression demonstrates that local linear instability is not required for self-oscillation: the necessary and sufficient condition is the global geometry—a folded slow manifold with two stable branches and a gate-controlled heating source exceeding the LPC threshold.

These results have direct implications for the design of autonomous gel oscillators. The decoupling of period (k), amplitude ($\mathcal{M}_g$ geometry), and waveform (n) provides three independent experimental handles. The subcritical topology implies that transitions into and out of the oscillatory state are hysteretic and abrupt, not gradual—a feature relevant to the implementation of gel-based mechanical switches and actuators. The insensitivity of the LPC boundary to gate function form suggests that oscillatory behavior can be expected across a broad class of LCST gel systems with collapse-modulated energy input, even when the phase transition is gradual rather than sharp.

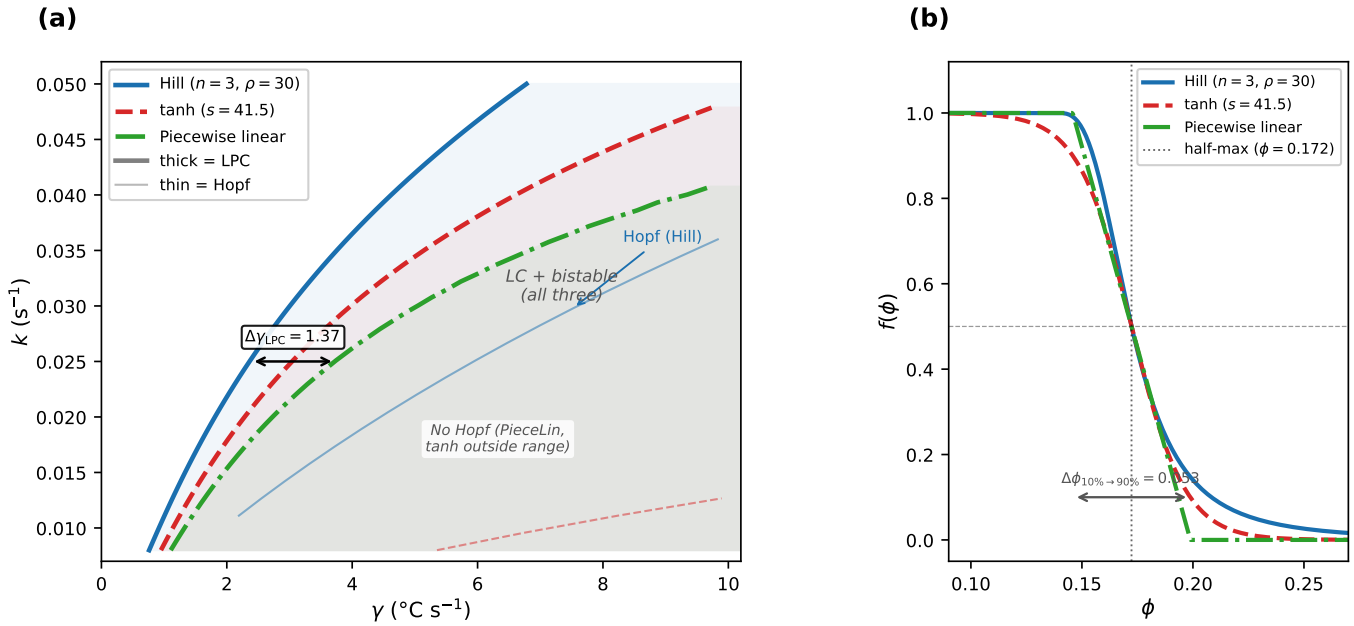

FIG. 7. Robustness of oscillation to gate function form. (a) LPC (thick curves) and Hopf (thin curves, where accessible) boundaries in the (γ, k) plane for three gate functions with equal transition width $\Delta\phi_{10\%\to90\%} = 0.053$: Hill (blue, solid), tanh (red, dashed), piecewise linear (green, dash-dot). The oscillatory region (right of the respective LPC curve) exists for all three forms. The double-headed arrow marks the LPC shift $\Delta\gamma_{\rm LPC} = 1.37\,^\circ\mathrm{C\,s^{-1}}$ at $k = 0.025\,\mathrm{s^{-1}}$. The thin blue curve (Hopf, Hill) lies inside the LPC boundary, confirming the subcritical structure. For the tanh gate, the Hopf boundary is visible only at the lowest k values (thin red segment); at $k = 0.025\,\mathrm{s^{-1}}$ it lies at $\gamma_{\rm H} \approx 30\,^\circ\mathrm{C\,s^{-1}}$, far outside the plot range. For the piecewise-linear gate no Hopf bifurcation was found for $\gamma < 120\,^\circ\mathrm{C\,s^{-1}}$: the fixed point remains stable throughout the oscillatory region and the limit cycle appears directly at the LPC. (b) Shapes of the three gate functions on the same ϕ axis; the bracket marks the equal transition width.

Appendix A: Equilibrium stability formulas

1. Closed-form derivatives

The stability of a fixed point depends on $g_\phi(\phi, T) = \partial g/\partial\phi$ [20–22]:

$$g_\phi = c_0 v_0 \left[\frac{1}{3\phi_0}\left(\frac{\phi}{\phi_0}\right)^{-2/3} - \frac{1}{2\phi_0}\right] + \left[1 - \frac{1}{1-\phi} + 2\chi_0\phi + 3\chi_1\phi^2\right]. \tag{A1}$$

Bistability vanishes at the cusp point defined by $g = g_\phi = g_{\phi\phi} = 0$, where

$$g_{\phi\phi} = c_0 v_0 \left[-\frac{2}{9\phi_0^2}\left(\frac{\phi}{\phi_0}\right)^{-5/3}\right] + \left[-\frac{1}{(1-\phi)^2} + 2\chi_0(T) + 6\chi_1\phi\right]. \tag{A2}$$

2. Mono/bistable boundary

The boundary separating monostable and bistable regimes in the $(c_0 v_0, \chi_1)$ plane is obtained by eliminating ϕ from $g(\phi, T) = 0$ and $g_\phi(\phi, T) = 0$. Define the auxiliary functions

$$\begin{aligned} E(\phi) &= \left(\frac{\phi}{\phi_0}\right)^{1/3} - \frac{\phi}{2\phi_0}, \\ E'(\phi) &= \frac{1}{3\phi_0}\left(\frac{\phi}{\phi_0}\right)^{-2/3} - \frac{1}{2\phi_0}, \\ A(\phi, T) &= \phi + \ln(1-\phi) + \chi_0(T)\phi^2, \\ B(\phi, T) &= 1 - \frac{1}{1-\phi} + 2\chi_0(T)\phi, \\ D(\phi) &= 3E(\phi) - \phi E'(\phi). \end{aligned} \tag{A3}$$

Solving the linear system $\{g = 0,\, g_\phi = 0\}$ for $(c_0 v_0, \chi_1)$ at fixed (ϕ, T) yields the parametric fold locus [Fig. 2(c)]:

$$c_0 v_0(\phi, T) = \frac{\phi B(\phi, T) - 3A(\phi, T)}{D(\phi)}, \tag{A4}$$

$$\chi_1(\phi, T) = \frac{E'(\phi)\, A(\phi, T) - E(\phi)\, B(\phi, T)}{\phi^2\, D(\phi)}. \tag{A5}$$

Sweeping ϕ over the physical range $(0, 1)$ traces the smooth boundary in Fig. 2(c).

Appendix B: Hopf normal form and Lyapunov coefficient

1. Linear stability details

The Jacobian matrix components require thermal derivatives. In the following normal-form calculation we linearize $\chi_0(T)$ about T_0, so that second and higher derivatives with respect to T are neglected within the considered temperature excursion.

$$g_T(\phi, T) \approx -\frac{\Delta h}{k_B T_0^2}\,\phi^2. \tag{B1}$$

For the Hill gate (9) with $x \equiv \max(\phi - \phi_g, 0)$, the derivative is:

$$f'(\phi) = \begin{cases} 0 & \phi \le \phi_g, \\ -\dfrac{n\kappa\,(\kappa x)^{n-1}}{\big(1+(\kappa x)^n\big)^2} & \phi > \phi_g. \end{cases} \tag{B2}$$

At the baseline $n = 3$ this gives $f'(\phi) = -3\kappa^3 x^2/(1+\kappa^3 x^3)^2$ for $\phi > \phi_g$, and $f'(\phi) = 0$ for $\phi \le \phi_g$.

2. Lyapunov coefficient formulas

We expand the vector field $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ around the Hopf equilibrium $\mathbf{x}^*$ as

$$\mathbf{F}(\mathbf{x}^* + \mathbf{u}) = J\mathbf{u} + \tfrac{1}{2}B(\mathbf{u},\mathbf{u}) + \tfrac{1}{6}C(\mathbf{u},\mathbf{u},\mathbf{u}) + \tfrac{1}{24}D(\mathbf{u},\mathbf{u},\mathbf{u},\mathbf{u}) + \cdots, \tag{B3}$$

where B, C, D are the symmetric multilinear forms of the second, third, and fourth derivatives of $\mathbf{F}$ at $\mathbf{x}^*$:

$$B_k(\mathbf{u},\mathbf{v}) = \sum_{i,j} \left.\frac{\partial^2 F_k}{\partial x_i \partial x_j}\right|_* u_i v_j,$$
$$C_k(\mathbf{u},\mathbf{v},\mathbf{w}) = \sum_{i,j,l} \left.\frac{\partial^3 F_k}{\partial x_i \partial x_j \partial x_l}\right|_* u_i v_j w_l. \tag{B4}$$

Let q be the right eigenvector of J with eigenvalue $i\omega_0$ and p the adjoint eigenvector satisfying $\langle p, q\rangle = 1$, where $\langle\cdot,\cdot\rangle$ denotes the standard Hermitian inner product. Define the auxiliary vectors

$$h_{20} = (2i\omega_0 I - J)^{-1}\,B(q,q),$$
$$h_{11} = -J^{-1}\,\mathrm{Re}\big[B(q,\bar{q})\big]. \tag{B5}$$

The first Lyapunov coefficient is then [27]

$$l_1 = \frac{1}{2\omega_0}\,\mathrm{Re}\big[\langle p, C(q,q,\bar{q})\rangle + 2\langle p, B(q,h_{11})\rangle + \langle p, B(\bar{q},h_{20})\rangle\big]. \tag{B6}$$

For the second Lyapunov coefficient l_2, additional auxiliary vectors are needed [27]:

$$h_{21} = (i\omega_0 I - J)^{-1}\big[-C(q,q,\bar{q}) - B(\bar{q},h_{20}) - 2B(q,h_{11}) + 2l_1 q\big],$$
$$h_{30} = (3i\omega_0 I - J)^{-1}\big[-C(q,q,q) - 3B(q,h_{20})\big]. \tag{B7}$$

The computation of l_2 then involves solving for h_{22} (via a projected singular system) and h_{31}, and projecting a fifth-order combination G_{32} onto the adjoint eigenvector [27]:

$$l_2 = \frac{1}{12\omega_0}\,\mathrm{Im}\langle p, G_{32}\rangle, \tag{B8}$$

where G_{32} collects all terms of the center-manifold expansion at order $z^3\bar{z}^2$ involving B, C, and D applied to q, $\bar{q}$, h_{ij}. All multilinear forms are evaluated numerically via symmetric finite differences.

3. l_1 and l_2 along the Hopf curve

Figure 3 shows l_1 evaluated at 19 points along the Hopf branch $\mathcal{H}$ over $\gamma_H \in [1.3, 11.6]$ (equivalently $k_H \in [0.007, 0.040]\,\mathrm{s}^{-1}$). No sign change is detected: l_1 remains strictly positive and monotonically decreasing throughout, ruling out a Bautin (generalized Hopf, $l_1 = 0$) codimension-2 point in this parameter window [27]. The subcriticality $l_1 > 0$ is consistent with the positive generalized-Hopf coefficient from AUTO-07p at every continuation point.

Because $l_2 > 0$ along the branch, the truncated normal form $\dot{r} = r(\mu + l_1 r^2 + l_2 r^4)$ has a very narrow validity range $\Delta\gamma \sim l_1^2/(4l_2)$ near each Hopf point. The global LPC branch therefore lies far outside this range and requires full numerical continuation via AUTO-07p [29].

Appendix C: Numerical methods

1. Simulation and classification

We integrate the ODEs using an adaptive Runge–Kutta method (SciPy `solve_ivp`). After discarding transients, the oscillation period is estimated from the autocorrelation of $\phi(t)$ on the late-time window: we compute the normalized autocorrelation $C(\tau)$ of the mean-subtracted signal and take the period as the lag of the dominant peak, $P_{\mathrm{est}} = \arg\max_\tau C(\tau)$. Basin maps are obtained by seeding initial conditions on a uniform grid in the (ϕ, T) plane and classifying each trajectory by its late-time behavior (fixed-point convergence or sustained oscillation).

2. Continuation settings

We use AUTO-07p [29] to track periodic orbit branches via orthogonal collocation. The LPC (Limit Point of Cycles) and BP (Branch Point) boundaries are identified by the test functions built into AUTO-07p, which monitor the non-trivial Floquet multipliers of the periodic orbit [27, 29]. Specifically: (i) an LPC occurs when a non-trivial Floquet multiplier passes through +1 (fold of cycles: stable and unstable periodic orbits coalesce); (ii) a BP occurs when a non-trivial Floquet multiplier passes through +1 at a branch-switching point where a secondary periodic-orbit family bifurcates from the primary branch. The oscillation boundary is defined by the outer envelope of $\mathcal{L}$ (LPC) and $\mathcal{B}$ (BP-connected) branches. Continuation is performed along one-parameter slices in γ at fixed k, and then two-parameter continuation is used to trace $\mathcal{L}$ and $\mathcal{B}$ as curves in the (γ, k) plane.

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